BSc 2024/2025

For internal students of Imperial College.

Mathematics Foundations (BUSI40001)

3 hours

XXXday ABth January 2024, 09:00 – 12:00

Instructions for the exam

- For Section A, internet access with ChatGPT is allowed and a computer with R or Python is needed.
- Section B is closed-book and paper-based. Section B is to made accessible to the students after completing Section A.
- Answer ALL questions (The mark breakdown is indicated throughout the exam).
- It is important to write neatly and clearly.
- If a question or part of a question is unclear to you, state any assumptions you need in order to complete the question.

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SECTION A (35 marks)

You have 45 minutes for Section A. Answer all **TWO** questions. Please submit your code. You must close your laptop after 45 minutes.

1. (20 marks) The ages at death of male parents and their male children are shown in the table below.

Person	A	B	C	D	E	F	G
adult	75	74	85	40	69	83	65
child	84	82	93	75	77	90	73

This question is about a male's lifespan given his father's lifespan.

(a) Write out the linear regression in the matrix form

$$\underline{y} = \mathbf{X}\underline{\beta} + \underline{\epsilon},$$

where $\underline{y}$, $\mathbf{X}$, $\underline{\beta}$, and $\underline{\epsilon}$ are to be defined. [2 marks]

(b) Show that the expression for $\underline{\beta}$ which minimizes $\underline{\epsilon}^\top \underline{\epsilon}$ is given by

$$\underline{\widehat{\beta}} = \left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T\underline{y}.$$

[5 marks]

(c) Show that the fitted values are given by

$$\underline{\hat{y}} = \mathbf{H}\underline{y}$$

where

$$\mathbf{H} \equiv \mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T$$

[3 marks]

(d) Show that $\mathbf{H}$ is idempotent and symmetric. [5 marks]

(e) Estimate the lifetime of a male whose male parent dies at age 89. You should do this using either R or Python [5 marks]

2. (15 marks: 5 marks for explanation, 10 marks for code)

Find and classify the critical points of the function $f(x, y) = x^2 + y^3 + xy$. Use code written in R or Python to carry out the necessary calculations, but explain the steps needed without showing algebraic details.

SECTION B (80 marks)

You have 2 hours 15 minutes for Section B. Answer all **FOUR** questions. Each question is worth 20 marks.

Question 1 (20 marks)

(a) Define what is meant by a *subsequence* of a sequence (a_n). [3 marks]

(b) Prove that if a sequence (a_n) converges to a limit L, then every subsequence also converges to L. [6 marks]

(c) Prove that every bounded sequence of real numbers has a monotone subsequence. [6 marks]

(d) State the Bolzano-Weierstrass Theorem. [2 marks]

(e) What is meant by the *peak* of a sequence? [3 marks]

Question 2 (20 marks)

(a) Define a *vector subspace* and give two examples in $\mathbb{R}^3$. [4 marks]

(b) Let U and W be subspaces of a vector space V. Define $U \cap W$ and $U + W$, and prove that both are subspaces of V. [6 marks]

(c) Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear transformation defined by $T(x, y) = (x + 2y, 3x - y)$. Find the matrix representation of T and determine its eigenvalues. [6 marks]

(d) Is T invertible? Justify your answer. [4 marks]

Question 3 (20 marks)

(a) Let A be the 3×3 matrix given by:

$$A = \begin{pmatrix} a & b & 0 \\ b & a & 0 \\ 0 & 0 & c \end{pmatrix},$$

where $a, b, c \in \mathbb{R}$ and $c \neq 0$. Find the characteristic polynomial of A. [5 marks]

(b) Determine the eigenvalues of A in terms of a, b, and c. [5 marks]

(c) For the case where $b \neq 0$, find the eigenvectors corresponding to each eigenvalue. [5 marks]

(d) Diagonalize the matrix A if possible, and explain your reasoning. [5 marks]

Question 4 (20 marks)

Consider the function $f(x, y) = x^2 + y^2$ subject to the constraint $g(x, y) = x + 2y - 4 = 0$.

(a) Set up the Lagrangian function $\mathcal{L}(x, y, \lambda)$ for this constrained optimization problem. [3 marks]

(b) Find the critical points by solving the system of equations given by the partial derivatives of the Lagrangian. [5 marks]

(c) Classify the critical points as local minima, maxima, or saddle points. [5 marks]

(d) Verify your result by substituting the critical points into $f(x, y)$ and interpreting the values. [4 marks]

(e) Explain the geometric interpretation of your solution. [3 marks]